How Blockchain Can Enhance Security and Trust in 5G/6G Network Slicing

Imagine a 5G/6G network serving billions of devices—including smartphones, autonomous vehicles, industries, IoT sensors, robots, drones, remote robotic surgeries, and smart city infrastructure—becoming interconnected at the same time. Each application requires a different level of latency, bandwidth, speed, reliability, and security. This is where network slicing becomes essential—but it also creates challenges. Ensuring secure communication, trust, privacy, and efficient data management in connected devices becomes increasingly challenging in 5G and 6G network slicing. Can blockchain provide that missing trust layer without becoming a bottleneck?

What is Network Slicing?

Network slicing is a 5G/6G technology that partitions a single physical network into multiple isolated virtual networks (called slices), each optimized with dedicated resources, security policies, bandwidth, latency, and Quality of Service (QoS) to meet the specific requirements of different applications such as healthcare, autonomous vehicles, and smart manufacturing. Network slicing is enabled by two key technologies:

Software-Defined Networking (SDN): Separates the network control plane from the data plane, allowing centralized and programmable network management.

Network Functions Virtualization (NFV): Replaces dedicated hardware appliances with virtual network functions running on standard servers.

Together, SDN and NFV make it possible to create, modify, and manage network slices efficiently. Although these slices share the same physical infrastructure, they must remain isolated, logically separated, and securely managed for different dedicated purposes.

Network Slice	Purpose	Requirements
Healthcare	Remote surgery	Ultra-low latency, high reliability
Autonomous Vehicles	Vehicle communication	Very low latency, high security
Smart Factory	Industrial IoT	High reliability, continuous connectivity
Video Streaming	Entertainment	High bandwidth
Smart City	IoT sensors	Massive device connectivity

The three primary categories of network slices are as follows:

1. eMBB (Enhanced Mobile Broadband)

Delivers extremely high-speed internet and large bandwidth. Its characteristics include very high download and upload speeds, latency between 10- 20 ms, and high throughput. Supports HD/4K/8K video streaming, high network capacity in crowded places, and improved user experience compared

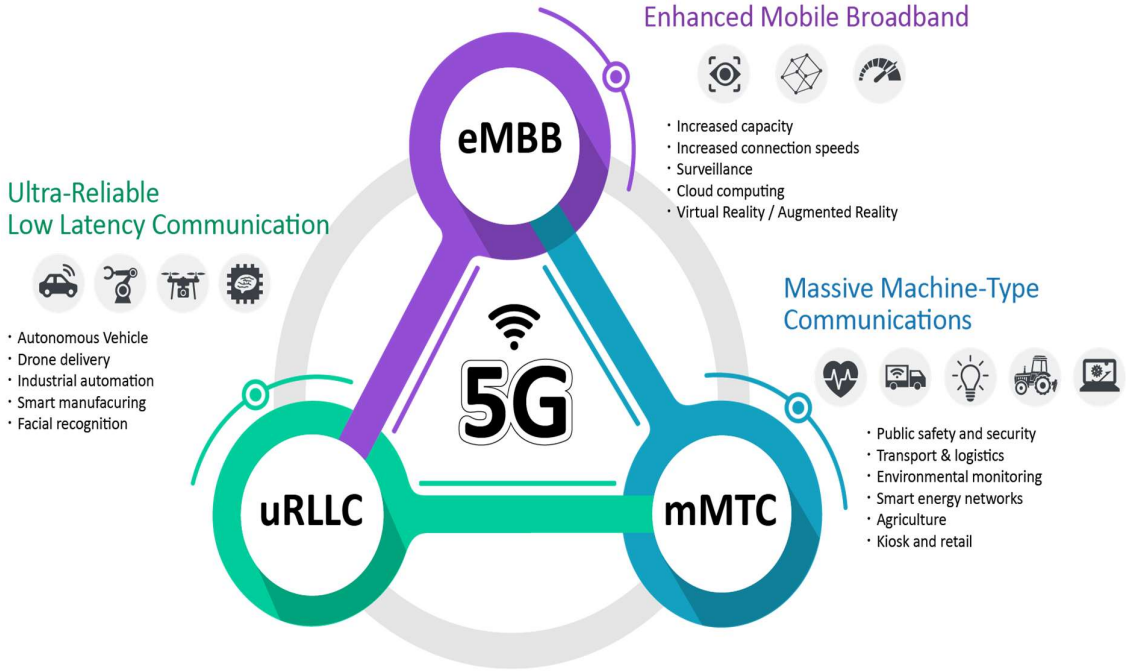
to 4G. Examples are Netflix 8K streaming, Virtual Reality (VR), Augmented Reality (AR), Cloud gaming, and high-speed mobile broadband.

2. URLLC (Ultra-Reliable Low-Latency Communications)

Enables applications where delays or failures are unacceptable. Its characteristics include ultra-low latency (around 1 ms), extremely high reliability (99.999% or higher), fast response time, and small packet transmission. Examples are Remote robotic surgery, Self-driving cars, Factory automation, Smart power grid protection, and drone control.

3. mMTC (Massive Machine-Type Communications)

Supports massive IoT deployments. Its characteristics include support for millions of connected devices, very low power consumption, small data packets, long battery life, and low cost. Examples are smart city sensors, smart electricity/water meters, agriculture monitoring, asset tracking, and environmental monitoring.



Current Security Challenges in Network Slicing

Because multiple organizations & tenants (customers) share the same infrastructure, several risks arise:

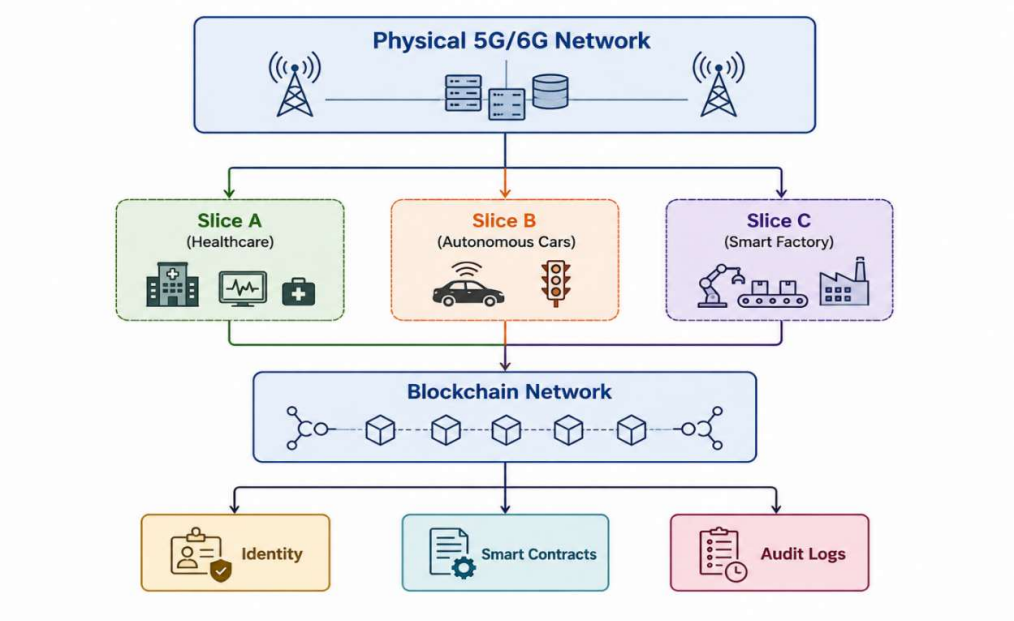
- **Unauthorized access to a slice** - Unauthorized users or attackers may gain access to a network slice and exploit its resources or sensitive data.
- **Data leakage between slices** - Improper isolation may allow confidential data from one network slice to be accessed by another slice.
- **Slice impersonation** - A malicious entity may pretend to be a legitimate network slice or service to deceive users or other network components.

- **Tampering with slice configurations** - Attackers may modify slice settings, such as security policies or QoS parameters, disrupting network performance or security.
- **Insider attacks** - Authorized users or administrators may intentionally misuse their privileges to steal data or compromise network slices.
- **Fraud in resource allocation** - Malicious users or operators may manipulate resource allocation to obtain unfair bandwidth, computing power, or spectrum resources.
- **Billing disputes** - Incorrect or manipulated usage records may lead to disagreements over service charges between network operators and customers
- **Single point of failure in centralized management** - If the centralized network controller fails or is attacked, it can disrupt the operation of all network slices.

Any unauthorized modification of network slices can have serious consequences, such as autonomous vehicle accidents or failures during remote robotic surgery. Blockchain helps address many of these challenges.

Blockchain-Based Network Slicing Architecture

Blockchain acts as a trusted management and security layer for all slices.



Blockchain Implementation

A permissioned management blockchain orchestration layer, such as Hyperledger Fabric, ensures that only authenticated and authorised stakeholders are participants. Blockchain eliminates reliance on a single controller for trust (Decentralization), prevents unauthorized modification of slice records (Immutability), Automates resource allocation, billing, and SLA enforcement (Smart Contracts), strengthens authentication for users and devices (Digital Identity), enables transparent investigations and compliance (Audit Trail), allows multiple stakeholders to verify shared operations and separate

billing (Transparency), detects unauthorized changes to configurations and logs (Data Integrity), reduces manual errors and administrative overhead (Automation).

1. Secure Slice Authentication

If authentication is weak, fake devices may join the slice, or hackers may gain access. So, each user, operator, and device has a blockchain-based digital identity. The authentication process consists of the following steps. The device requests access → Blockchain verifies its identity → Smart contract checks permissions → Access is granted only if authorized →The authentication event is permanently recorded. Suppose a hospital's surgical robot requests access to the healthcare slice. Blockchain verifies: Robot identity, Hospital authorization, Valid digital certificate; only then is access granted.

2. Secure Resource Allocation Using Smart Contracts

Network slices require resources such as bandwidth, CPU, Memory, Storage, and radio spectrum. Traditionally, allocation is managed by centralized controllers. Blockchain can automate allocation using smart contracts. Suppose a hospital reserves 200 Mbps bandwidth and 2 ms latency. The smart contract verifies and records the resource-allocation agreement, while the network slice orchestrator performs the actual resource allocation. The allocation process becomes automated, transparent, and auditable, reducing manual intervention and enabling customer requirements to be fulfilled more efficiently and transparently.

3. Immutable Slice Configuration

Each slice maintains configuration details. If an attacker changes these settings (Quality of Service, bandwidth, resources, security policies, routing rules), service quality or security may be compromised. Blockchain stores cryptographic hashes of approved configurations. If a configuration changes unexpectedly, the hash no longer matches, and the system detects tampering immediately.

4. Isolation Between Slices

A cyberattack on one slice should not affect others. An attack on the entertainment slice cannot change records or permissions for the healthcare slice because each slice's management records are independently verified. Blockchain supports slice isolation by maintaining tamper-evident access permissions, resource assignments, and security policies, while actual network isolation is enforced by SDN/NFV and the network slice orchestrator.

5. Secure & Trustless Data Sharing

'Trustless' means that participants do not need to trust a central authority because trust is established through blockchain consensus and cryptographic verification. So, multiple telecom operators and service providers may jointly manage slices and do business transparently. Blockchain provides a shared, trusted ledger where all authorized participants can verify resource usage, configuration updates, service-level agreements (SLAs), and access events. This reduces disputes and improves collaboration.

6. Audit Trail and Accountability

Every important operation is recorded on the blockchain, including slice creation, user authentication, configuration changes, resource allocation, and policy updates. Because these records are tamper-evident, administrators can determine who has performed an action, when it happened, and what was changed. This is valuable for security investigations and regulatory compliance.

7. SLA (Service Level Agreement) Enforcement

Service Level Agreements (SLAs) define the contractual performance guarantees for each network slice. Smart contracts can continuously evaluate performance metrics. Suppose in the agreed SLA, Latency < 5 ms & Availability = 99.999%. The network monitoring system/AI agent should measure latency, availability, bandwidth, etc.; the blockchain records the result, and the smart contract executes the contractual action. If the SLA is violated, a violation is recorded, notifications are generated, and penalties or compensation can be triggered automatically, depending on the agreement.

8. Billing and Charging

Blockchain enables accurate, transparent, and tamper-resistant billing, even when multiple service providers share the same infrastructure. Each slice's usage is monitored and recorded. Network monitoring systems collect bandwidth usage, time consumed, and data transferred, while blockchain securely records these events to ensure transparent and tamper-resistant billing. Smart contracts can automatically calculate charges and settle payments, reducing fraud and reconciliation effort by recording each transaction.

Scenario in Robotic Surgery

Suppose a hospital performs remote robotic surgery.

Step 1: The surgeon and robot log in using a blockchain-based digital identity.

Step 2: The hospital requests the healthcare slice(URLLC) with high bandwidth and ultra-low latency.

Step 3: The smart contract can verify authorization.

Step 4: Bandwidth and low-latency resources are reserved & further verified using Blockchain.

Step 5: Configuration hashes ensure no one has altered slice settings.

Step 6: All access, resource allocation, and policy changes are logged.

Step 7: If an attacker attempts to modify slice policies, the blockchain detects the mismatch and administrators are alerted.

We do not use blockchain for packet-level or real-time network operations. We use a permissioned Hyperledger Fabric blockchain as a trusted management-plane layer for authentication,

authorization, configuration integrity, SLA management, auditability, and billing. The slice orchestrator and SDN/NFV infrastructure perform the actual real-time resource allocation and isolation.

Future advancements with AI & Post-Quantum Cryptography

Current blockchain systems primarily rely on public-key cryptographic algorithms such as RSA, ECC, and ECDSA. These algorithms could be compromised by sufficiently powerful quantum computers using Shor's algorithm. To address this future threat, blockchain systems are expected to adopt Post-Quantum Cryptography (PQC), including CRYSTALS-Kyber for key encapsulation and CRYSTALS-Dilithium for digital signatures. Furthermore, AI-based intrusion detection and intelligent slice management can automatically detect cyberattacks and dynamically update blockchain-based access control policies.

Conclusion

Blockchain enhances the security of 5G and future 6G network slicing by providing trusted identities, tamper-evident records, automated, transparent resource management via smart contracts, secure auditing & billing, and transparent collaboration among multiple operators. Rather than replacing existing telecom technologies, blockchain serves as a decentralised trust & security layer that protects slice management and supports secure, reliable, and accountable operation of next-generation mobile networks, especially for critical applications such as healthcare, autonomous transportation, industrial automation, and smart cities. In the future, the integration of Blockchain, Artificial Intelligence, Post-Quantum Cryptography, and 5G/6G network slicing is expected to create a secure, intelligent, and autonomous networking architecture capable of addressing both current and emerging cyber threats.

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